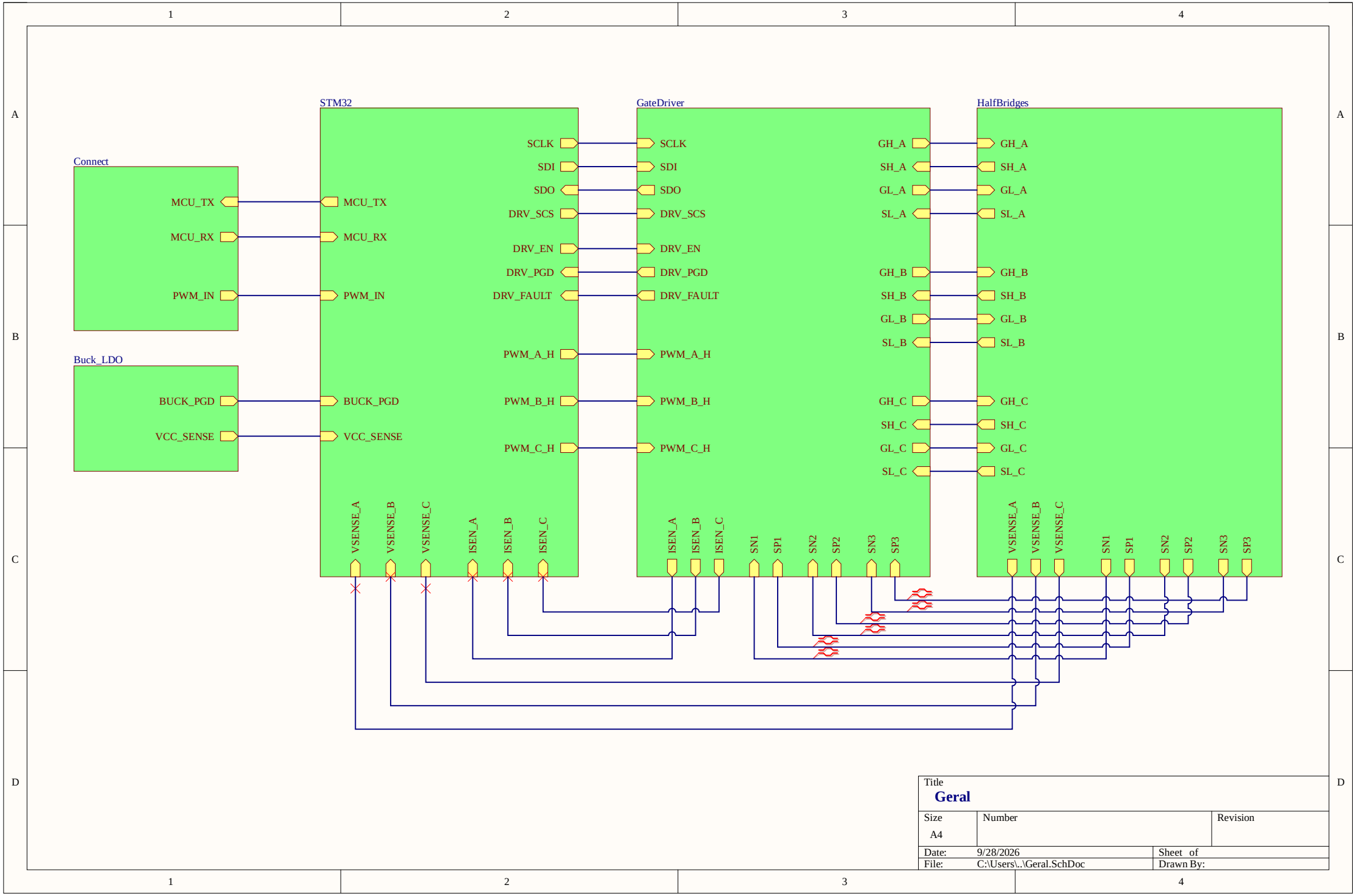
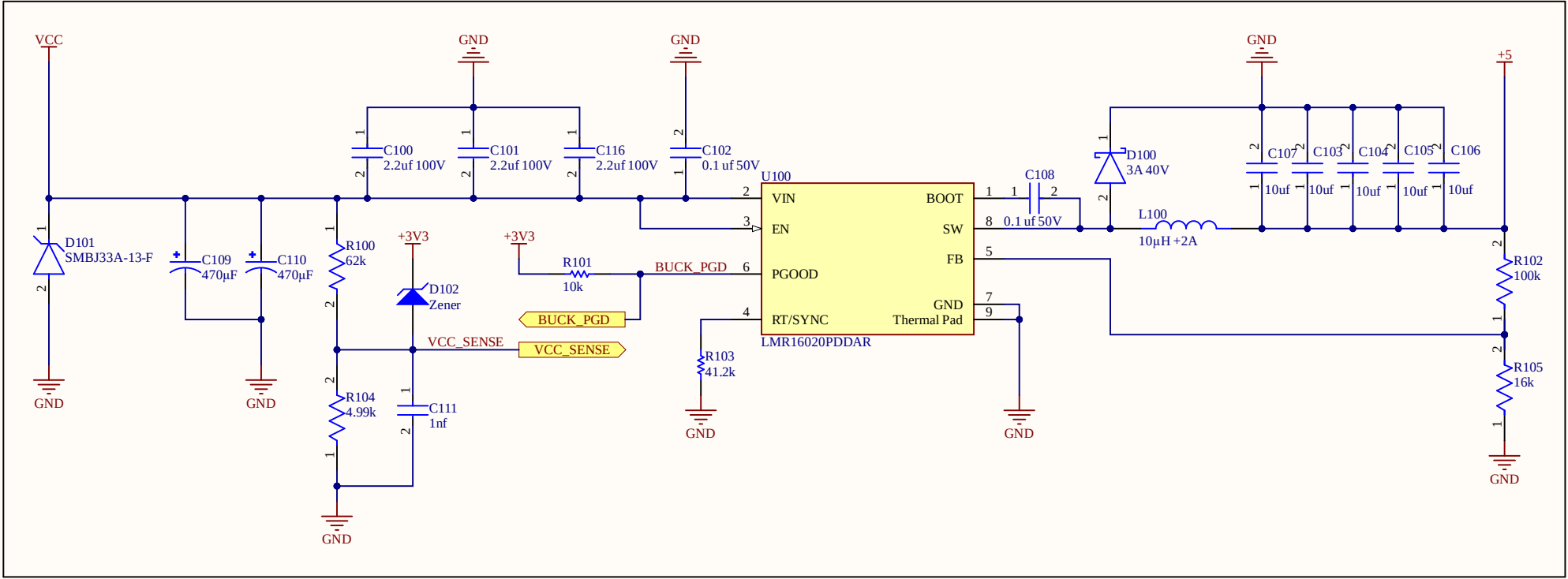


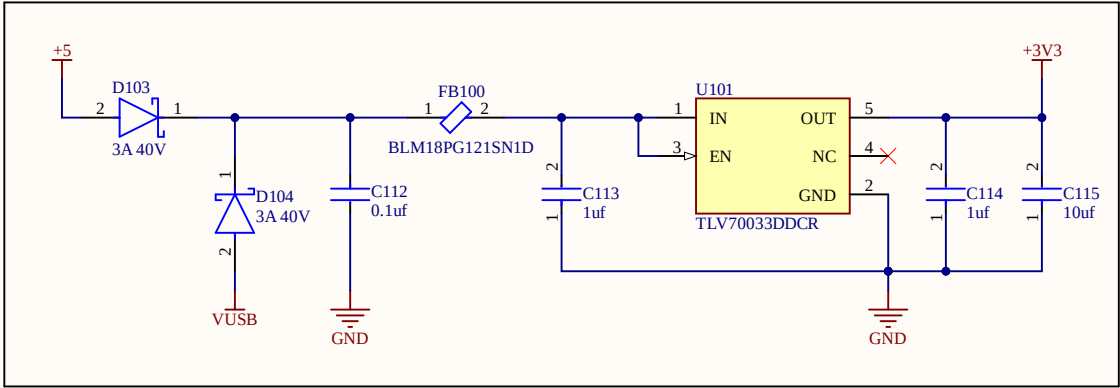


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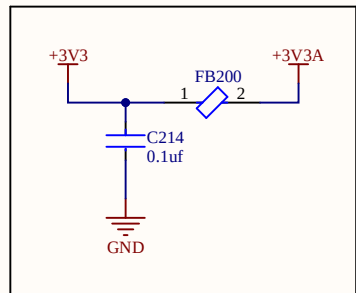
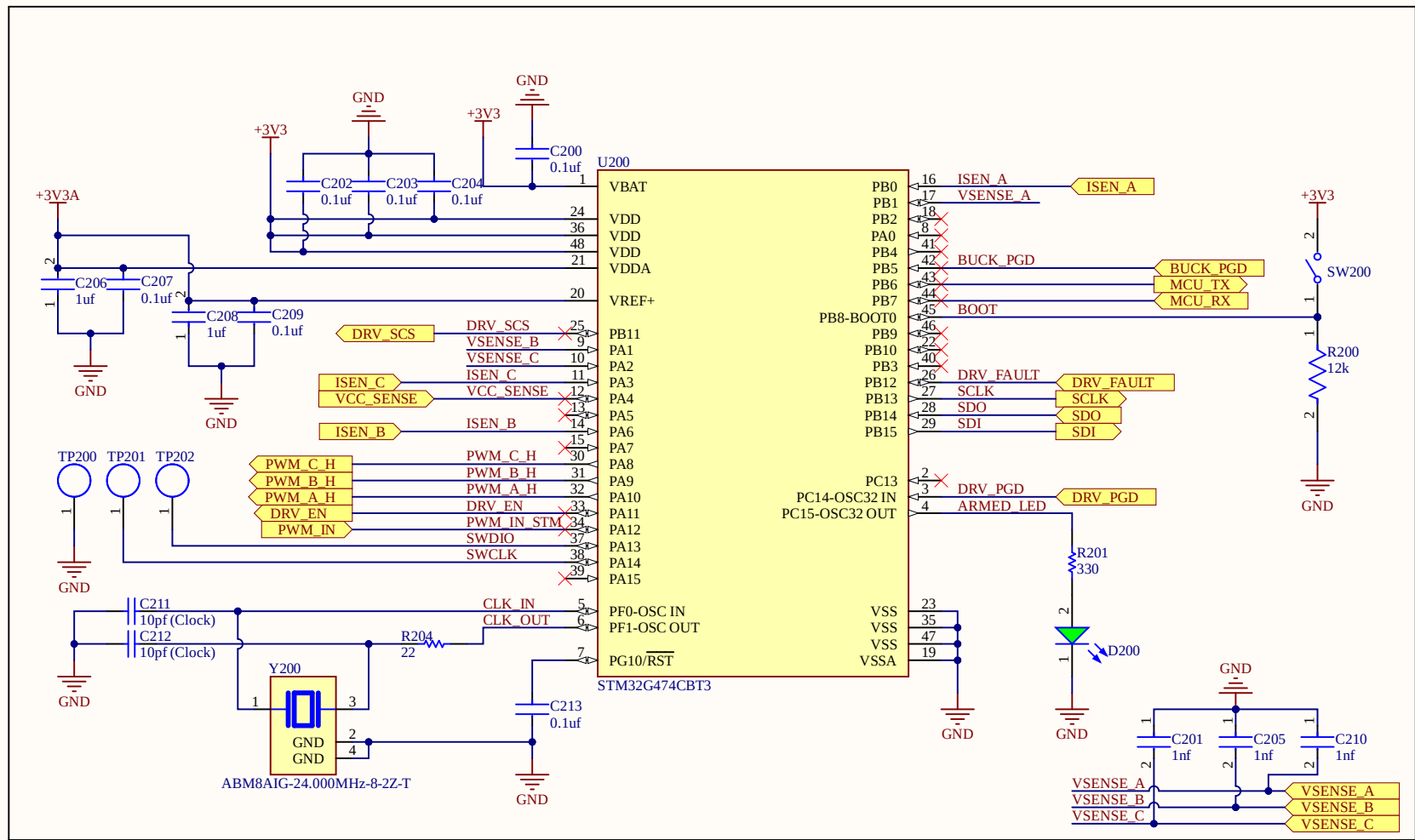
2S-6S to 5V



5V to 3V3

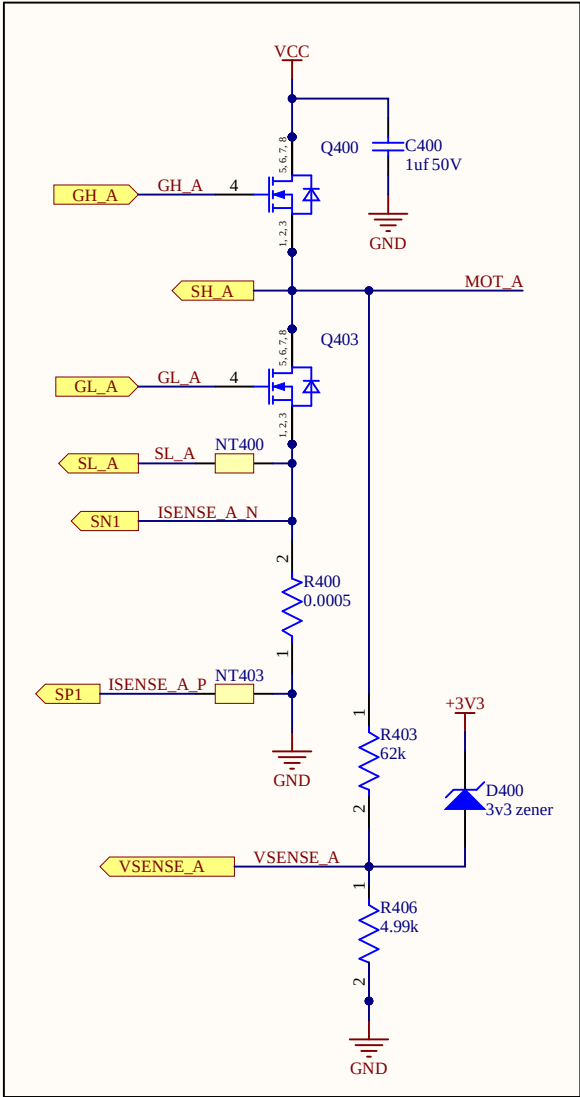


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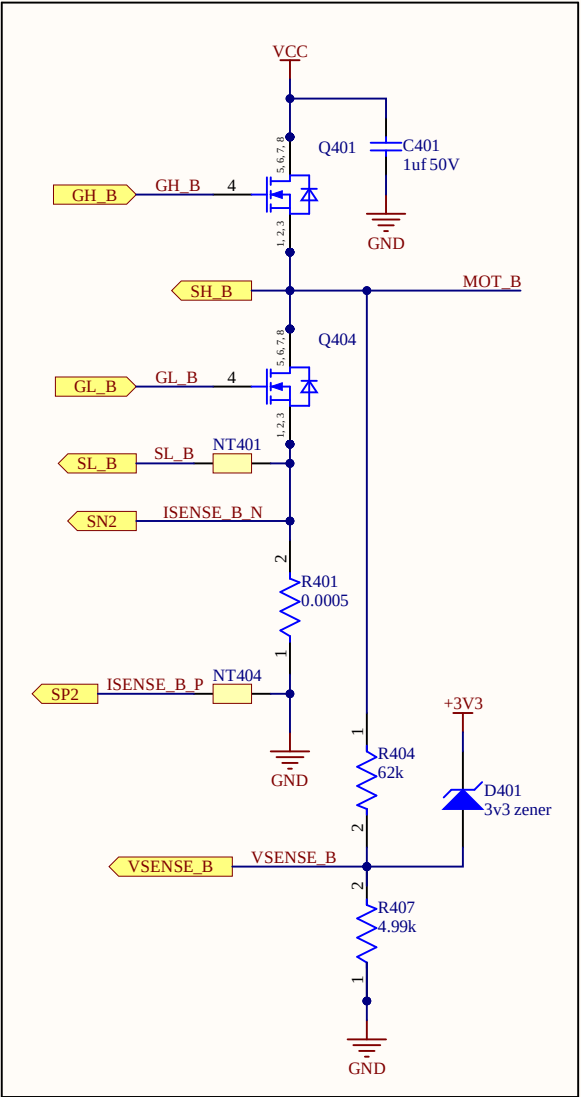


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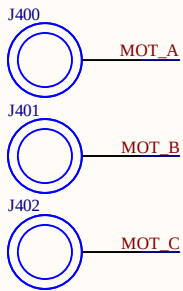
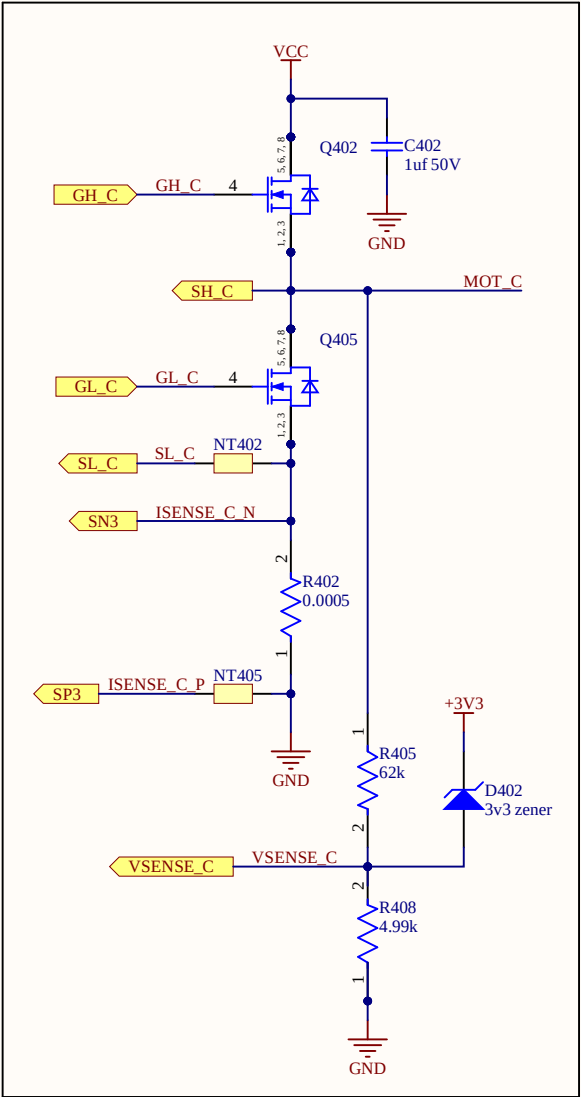
Phase A



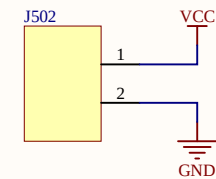
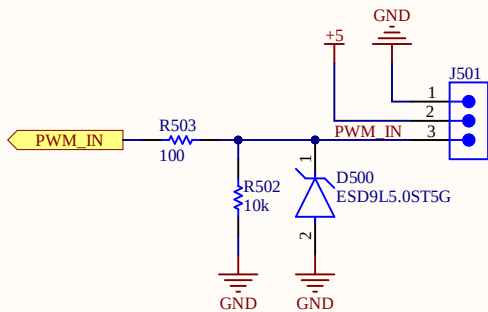
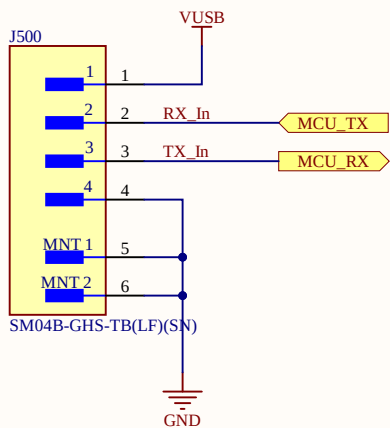
Phase B



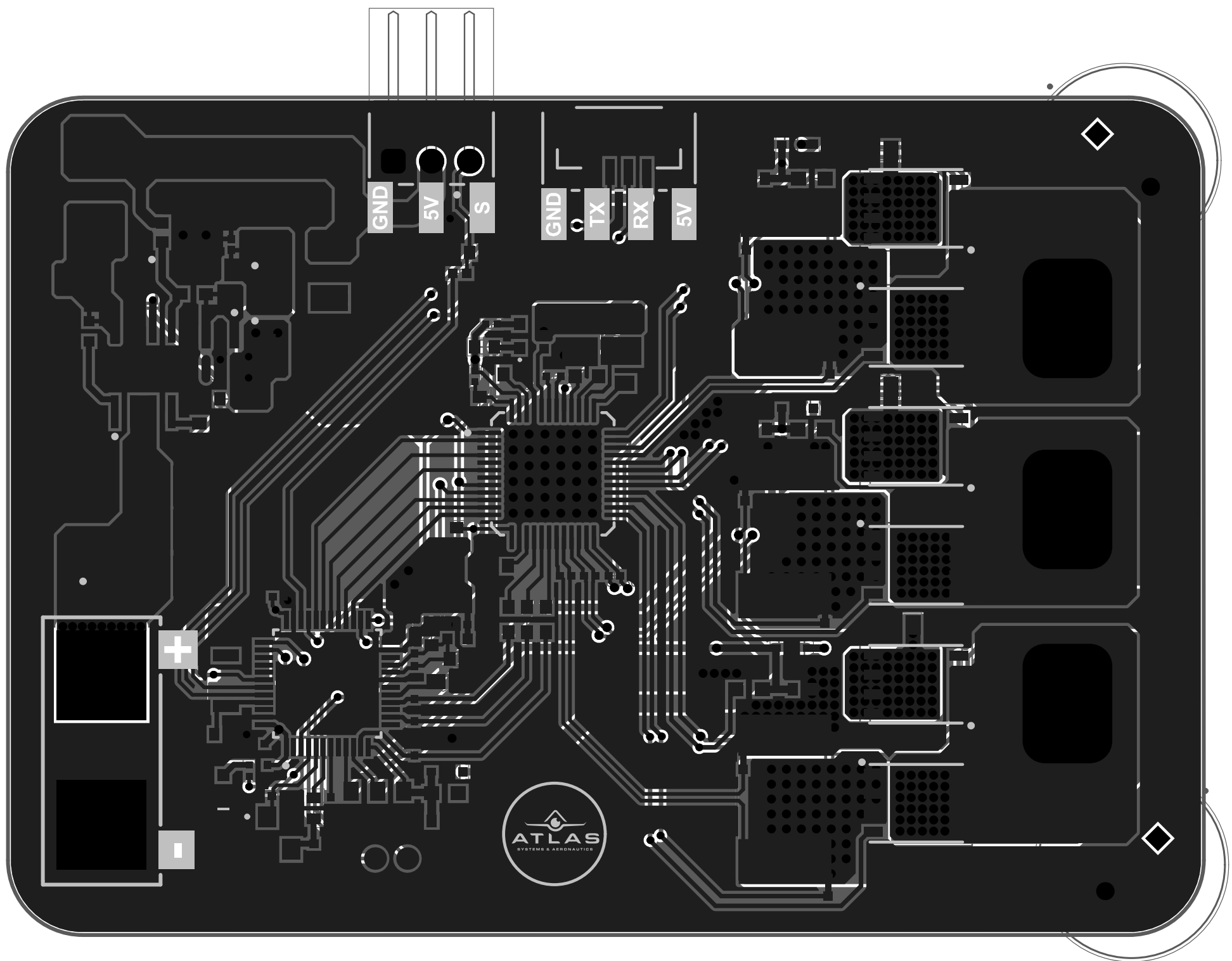
Phase C



Title		
Half-Bridges e Back-EMF Sense + Conector Output		
Size	Number	Revision
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Date:	9/28/2026	Sheet of
File:	C:\Users\...\HalfBridges.SchDoc	Drawn By:



Title		
Conectores Input (UART + Bateria + SR)		
Size	Number	Revision
A4		
Date:	9/28/2026	Sheet of
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Designator	Name	Description	Revision ID	Line #	Revision Status	Quantity	Manufacturer 1	Manufacturer Part Number 1	Manufacturer Lifecycle 1	Supplier 1	Supplier Part Number 1	Supplier Unit Price 1	Supplier Subtotal 1
C100, C101, C116, C303	GCJ32DR72A225KA01L	CAP CER2.2uF 100V X7R 1210	CMP-06035-003757-1		Up to date	4	Murata	GCJ32DR72A225KA01L	Volume Production	Arrow Electronics	GCJ32DR72A225KA01L	1.19	4.74
C102, C108	GCJL88R71H104MA12J	Cap Ceramic 0.1uF 50V X7R 20% Pad SMD 0603 Soft Termination +125°C Automotive T/R	CMP-06035-022661-1		Up to date	2	Murata	GCJL88R71H104MA12J	Volume Production				
C103, C104, C105, C106, C107, C115	C1206C106K4RACAUTO	CAP CER10UF 16V X7R 1206	CMP-03020-003375-1		Up to date	6	Yageo Group	C1206C106K4RACAUTO	Volume Production	Arrow Electronics	C1206C106K4RACAUTO	0.18768	1.88
C109, C110	EBJFC1H471	Condensateur, alu/nlrique, 470 uF, Tol:±Grance 20%, Vol-Rtg 50 VCC, Radial, 12,5 x25, Bulk				2							
C111, C201, C205, C210	CC0603JNFP09BN102	CAP CER1000PF 50V COG/NPO 0603	CMP-03422-001494-1		Up to date	4	Yageo Group	CC0603JNFP09BN102	Volume Production	Arrow Electronics	CC0603JNFP09BN102	0.01167	0.04668
C112, C200, C202, C203, C204, C207, C209, C213, C214	CL05B104KA6NNNC	Cap Ceramic 100nF 25V X7R±10% SMD 0402 +125°C Paper T/R	CMP-13271-001643-1		Up to date	9	Samsung Electro-Mechanics	CL05B104KA6NNNC	Volume Production	Mouser	187-CL05B104KA6NNNC	0.02139	0.21393
C113, C114, C206, C208, C300, C301, C313	CC0603K9P7R7BB105	CC0603 1 uF X7R 10.00% 16 V	CMP-03422-001047-1		Up to date	7	Yageo Group	CC0603C105K3RAC1U	Volume Production	Newark	CC0603C105K3RAC1U	0.33257	3.33
C211, C212	GRM1555C1H100JA01D		CMP-2008-04245-1		Up to date	2	Murata	GRM1555C1H100JA01D	Not Recommended for New Design	Arrow Electronics	GRM1555C1H100JA01D	0.01206	0.12058
C302, C400, C401, C402	GRM21BR71H105KA12L		CMP-1036-04761-2		Up to date	4	Murata	GRM21BR71H105KA12L	End of Life	Arrow Electronics	GRM21BR71H105KA12L	0.07235	217.04
C304	GRM31CR71H475KA12L	Chip Multilayer Ceramic Capacitors for General Purpose, 1206, 4.7uF, X7R, 15%, 10%, 50V	CMP-1037-04923-3		Up to date	1	Murata	GRM31CR71H475KA12L	Volume Production	Amet	GRM31CR71H475KA12L	0.07267	145.34
C305	CL10B473KCBWPN	0.047 uF ±10% 100V Ceramic Capacitor X7R0603 (1608 Metric)				1							
C306	CC0402K9X7R9BB473	CAP CER0.047uF 50V X7R0402	CMP-03422-000113-1		Up to date	1	Yageo Group	CC0402K9X7R9BB473	Volume Production	Arrow Electronics	CC0402K9X7R9BB473	0.09287	0.09287
C307, C308, C309	CL05B102KB5NNNC	Cap Ceramic 1nF 50V X7R±10% SMD 0402 +125°C Paper T/R	CMP-13271-004091-1		Up to date	3	Samsung Electro-Mechanics	CL05B102KB5NNNC	Volume Production	DigKey	1276-1032-6-ND	0.02723	0.27228
C310, C311, C312	CL05C102JB5NNNC	Cap Ceramic 1nF 50V COG±5% SMD 0402 +125°C Paper T/R	CMP-13271-002012-1		Up to date	3	Samsung Electro-Mechanics	CL05C102JB5NNNC	Volume Production	Newark	CL05C102JB5NNNC	0.09919	0.29756
D100, D103, D104	B340A-13-F	DIODE SCHOTTKY 40V 3A SMA	CMP-1667-00003-3		Up to date	3	Diodes Inc.	B340A-13-F	Volume Production	Arrow Electronics	B340A-13-F	0.39422	1.18
D101	SMB33A-13-F	TVS DIODE 33V 53.3V SMB	CMP-12828-000031-1		Up to date	1	Diodes Inc.	SMB33A-13-F	Volume Production	Arrow Electronics	SMB33A-13-F	0.52248	0.52248
D102, D400, D401, D402	BZT52C3VS-7-F	Surface Mount Zener Diode, 3.3 V, -65 to 150 degC, 2-Pin SOD323, RoHS, Tape and Reel	CMP-2000-05975-1		Up to date	4	Diodes Inc.	BZT52C3VS-7-F	Volume Production	Amet	BZT52C3VS-7-F	0.02678	80.34
D200	150060GS75000	430nm Colorless transprence 520nm Emerald 140° LED0603 Light Emitting Diodes (LED) ROHS	CMP-1426-00002-4		Out of date	1	Würth Elektronik	150060GS75000	Volume Production	DigKey	732-4971-6-ND	0.14586	0.14586
D300	150060R55040	LED RED DIFFUSED 0603 SMD	CMP-207942-000007-3		Up to date	1	Würth Elektronik	150060R55040	Volume Production	DigKey	732-12015-1-ND	0.18476	0.18476
D500	ESD9L5.0S15G		CMP-1063-00114-2		Up to date	1	onsemi	ESD9L5.0S15G	Volume Production	Arrow Electronics	ESD9L5.0S15G	0.05222	0.05222
FB100	BLM18PGL121SN1D	Chip Ferrite Bead, 0603, 120Q @100MHz, 0.05Q, 25%, 2A	CMP-0686-00557-7		Up to date	1	Murata	BLM18PGL121SN1D	Volume Production	Arrow Electronics	BLM18PGL121SN1D	0.09287	0.09287
FB200	BLM18AG601SN1D	Chip Ferrite Bead, 0603, 600Q @100MHz, 0.38Q, 25%, 500mA	CMP-0686-00436-3		Up to date	1	Murata	BLM18AG601SN1D	Volume Production	Mouser	81-BLM1LA601S	0.09724	0.09724
J500	SM04B-GHS-TB(LF)(SN)	CONN HEADER SMD RA 4POS 1.25MM	CMP-17439-000411-1		Up to date	1	JST	SM04B-GHS-TB(LF)(SN)	Volume Production	Mouser	306-SM04BGHSTBLSFN	0.37924	0.37924
J501	61300311021	THT Angled Pin Header WR-PHD, Pitch 2.54 mm, Single Row, 3 pins	CMP-1502-01061-4		Up to date	1	Würth Elektronik	61300311021	Volume Production	DigKey	732-5336-ND	0.36952	0.36952
L100	MMSV06035-100MT					1							
Q400, Q401, Q402, Q403, Q404, Q405	BSC014N06NS	OptiMOS Power - MOSFET, 60 V				6	Infineon	BSC014N06NSATMA1	Volume Production	Newark	BSC014N06NSATMA1	3.05	18.3
R100, R403, R404, R405	CPF0603B62KE1	RES SMD 62KQ 0.1% 1/16W 0603	CMP-03211-004922-1		Up to date	4	TE Connectivity	CPF0603B62KE1	Volume Production	Mouser	279-CPF0603B62KE1	0.29173	1.17
R101, R502, R504, R502	ER3BFNF1002V	Chip Resistor, 10 KOhm, +/- 1% 0.1 W, -55 to 155 degC, 0603 (1608 Metric), RoHS, Tape and Reel	CMP-1012-00623-3		Up to date	4	Vishay	CRCW060310K0RKEC	Volume Production	Arrow Electronics	CRCW060310K0RKEC	0.00107	0.00428
R102	CRGCO0603JL00K	CRGCO 0603 100K 5%	CMP-03211-009800-1		Up to date	1	TE Connectivity	CRGCO0603JL00K	Volume Production	Arrow Electronics	CRGCO0603JL00K	0.01566	0.01566
R103	CRCW0603412K9FEA		CMP-2000-03338-1		Up to date	1	Vishay	CRCW0603412K9FEA	Volume Production	Arrow Electronics	CRCW0603412K9FEA	0.00107	0.00107
R104, R406, R407, R408	AF0603FR074K99L	Anti-Sulfurated Chip Resistors Automotive Grade 4.99kQ 1% 100ppm/°C 0603	CMP-03412-046861-1		Up to date	4	Yageo Group	AF0603FR074K99L	Volume Production	Mouser	603-AF0603FR074K99L	0.02139	0.21393
R105	CRCW060316K0RFEA		CMP-1012-00644-3		Up to date	1	Vishay	CRCW060316K0RFEA	Volume Production	Arrow Electronics	CRCW060316K0RFEA	0.00049	0.00049
R200	AC0603FR1012HL	RES Thick Film, 16kQ, 1% 0.1W, 100ppm/°C, 0603	CMP-03412-031259-2		Up to date	1	Yageo Group	AC0603FR1012HL	Volume Production	DigKey	13-AC0603FR1012KLDNR-ND	0.09724	0.09724
R201, R301, R303	RC0402FR07330RL		CMP-2002-07613-1		Up to date	3	Yageo Group	RC0402FR07330RL	Volume Production	Mouser	603-RC0402FR07330RL	0.009	0.09
R204	RC0603FR0722JL	Chip Resistor, 22 Ohm, +/- 1% 0.1 W, -55 to 155 degC, 0603 (1608 Metric), RoHS, Tape and Reel	CMP-1659-00033-1		Up to date	1	Yageo Group	RC0603FR0722JL	Volume Production	Arrow Electronics	603-RC0603FR0722JL	0.04979	0.04979
R303, R503	RC0402FR07100RL		CMP-2002-07344-1		Up to date	2	Yageo Group	RC0402FR07100RL	Volume Production	Mouser	603-RC0402FR07100RL	0.00872	0.09724
R305, R306, R307	RC0603FR0747JL		CMP-1659-00039-2		Up to date	3	Yageo Group	RC0603FR0747JL	Volume Production	Mouser	603-RC0603FR0747JL	0.0107	0.10697
R400, R401, R402	PL3921JNPF50J5L	METAL SHUNT RESISTOR 3921 5% 175	CMP-03412-005804-1		Up to date	3	Yageo Group	PL3921JNPF50J5L	Volume Production				
SM200	B3U-1000P	SWITCH TACTILE SPST-NO 0.05A 12V	CMP-2100-03586-2		Up to date	1	Omron	B3U-1000P	Volume Production	RS (Formerly Allied Electronics)	B3U-1000P	0.72932	0.72932
U100	LMR16020PDDAR	IC REG BUCK ADJUSTABLE 2A RSOPWR	CMP-04918-000084-1		Up to date	1	Texas Instruments	LMR16020PDDAR	Volume Production	Mouser	595-LMR16020PDDAR	4.35	4.35
U101	TLV70033DDCR	IC REG LINEAR 3.3V 200MA SOT23-5	CMP-0394-01056-3		Up to date	1	Texas Instruments	TLV70033DDCR	Volume Production	Arrow Electronics	TLV70033DDCR	0.07663	0.07663
U200	STM32G431CBT6	MAINSTREAM ARM CORTEX-M4+ MCU 17	CMP-12105-000030-1		Up to date	1	STMicroelectronics	STM32G474CBT3	Volume Production	Mouser	511-STM32G474CBT3	8.71	8.71
U300	DM8305NEHFQ1	IC MTRDRV MULTI 4.4-45V 48HTQFP	CMP-04915-000021-1		Up to date	1	Texas Instruments	DM8305NEHFQ1	Volume Production				
Y200	ABMBAIG-24.000MHz-8-22-T	Crystal 24MHz ±20ppm (Tol) ±50ppm (Stability) 8pF FUND 50Ohm Automotive 4-Pin Ultra Mini-CSMD T/R				1	Abracon	ABMBAIG-24.000MHz-8-22-T	Volume Production	Newark	ABMBAIG-24.000MHz-8-22-T	0.58911	0.58911